

Context-Aware Phishing: Leveraging Transformer and BiLSTM Networks for Cybersecurity

S. Vasudevan

Department of Mathematics
Hindustan Institute of Technology and
Science
Padur, Chennai
svasudevanidvrs@gmail.com

Vediyappan Govindan

Department of Mathematics
Hindustan Institute of Technology and
Science, Padur, Chennai.
govindoviya@gmail.com

Abstract— Background: Phishing attempts now pose significant cybersecurity threats with their intricacy. Conventional machine learning models sometimes struggle to uncover deep contextual and sequential patterns from email content, therefore producing poor detection results. Deep learning approaches as Transformers and BiLSTMs have emerged as practical substitutes to address these challenges. **Novelty & Originality:** This work provides a hybrid deep learning model to improve phishing email classification by aggregating a Transformer encoder with a BiLSTM network. Our model uses BiLSTMs for learning sequential dependencies and multi-head attention mechanisms for contextual feature extraction unlike conventional approaches. Furthermore, used to improve generalization are L2 regularization and dropout strategies. The dataset consists of both phishing emails produced generatively and authentically, hence strengthening the resilience of the algorithm. **Conclusion:** Ultimately, testing results show that our proposed Transformer-BiLSTM model achieves an extraordinary 99% accuracy, much above accepted methods. This extraordinary accuracy emphasizes how relevant it is for email security systems. Through efficient capture of contextual and sequential patterns, our approach enhances phishing detection capabilities, so boosting cybersecurity solutions.

Keywords— Deep Learning, Synthetic Emails, Neural Architectural Search, Transformers, Python

I. INTRODUCTION

One of the leading cybersecurity problems that continues to exist to this day are phishing attacks, which exploit human vulnerabilities to gain unauthorized access to valuable information. The proliferation of sophisticated cybercrime strategies may render the results of traditional email classification unable to fully differentiate a phishing message from a legitimate one. The recent progress in the field of artificial intelligence (AI) has given an impulse to the massive-scale production of synthesized material such as deepfake movies, AI-based text, or data-altered images, inevitably triggering widespread anxieties regarding misinformation, deception, and moral integrity [4]. For consumers, companies and regulatory agencies, the more realistic AI-generated content becomes, separating real from fake has become a far more daunting challenge.

Deepfake technology has made the creation of highly realistic synthetic media substantially easier with the help of generative adversarial networks (GANs) and diffusion models [1]. AI-generated content is being misappropriated for fraudulent reasons like malicious disinformation, fabricated testimony, and industrial espionage (Mishra et al., 2025), albeit being used for well applications like entertainment, academic research, and digital marketing.

Vision language-based deepfake detection techniques, and linguistic cues that reveal the differences between the

generated and real text formed the part of the detection techniques (Lekshmi Ammal, 2025). To enhance detection accuracy, some scholars have studied multimodal detection systems based on image, text and metadata analysis [15]. However despite these advances, adversarial methods have continued to evolve and circumvent current detection systems and require more effective counter measures (Ain et al., 2025).

A. Approach Overview

In this paper, we propose a novel hybrid deep learning for phishing emails classification integrating Transformer and BiLSTM. Conventional methods fail to effectively mine important contextual and sequential relations inside an email corpus, and they often increase a high rate of missing detection cases. The proposed method borrows multi-head attention from Transformers to capture context information and further incorporates Bidirectional LSTMs (BiLSTMs) to model the dependence in the data. Regularization methods, including L2 regularization and dropout, are also used to reduce overfitting and improve generalization. The data set contains both real and AI-generated phishing emails to make them more robust.

II. LITERATURE REVIEW

Many studies highlight how well machine learning approaches find phishing emails. Evaluating several categorization models including Support Vector Machines (SVM), Navie Bayes (NB), and Long Short- Term Memory (LSTM), Ajay, Garg, and Prasad (2024 [2]) scored accuracy rates ranging from 99.62% to 99.62%. Still, these systems sometimes rely on manually created features, which might not be able to broadly apply to the always growing phishing techniques.

We will go over the existing research on detecting phishing emails in a systematic way by dividing it into four main groups: (i) traditional machine learning models, (ii) deep learning architectures like LSTM and BiLSTM, (iii) Transformer-based models, and (iv) hybrid approaches that use both contextual and sequential modelling. This organisation shows how tactics have changed throughout time and helps us see the pros and cons of each group. Phishing email language aspects have been investigated using NLP approaches.

Examining advanced NLP algorithms for phishing detection, Salloum et al. (2021) [14] underlined their importance in identifying complex phishing attempts. Notwithstanding advances, phishing attacks using AI-generated text challenge current NLP algorithms due to their human-like coherence and contextual relevance.

Vulnerability to phishing is influenced by human cognitive distortions. Researchers Wang et al. (2012) looked at how people responded to phishing emails and found that

detection depends on paying close attention to visual signals and deception markers. Bengio, Y.; Simard, P.; Frasconi, P [12] noted that phishing success rates are much reduced by user knowledge and education. Still, phishing emails produced by artificial intelligence could more deftly use psychological manipulation and emotional triggers than more traditional phishing methods.

Alzhrani, K et al. [5] underlined the growing complexity of phishing attacks using artificial intelligence to produce quite convincing phishing material. These attacks go past

dataset. It provides a useful foundation for cybersecurity research by combining real phishing techniques with accepted email patterns, therefore enabling the building of strong defenses against email-related assaults.

B. Dataset Composition

Phishing emails and genuine emails make up the two main groups in the dataset. Often using urgent subject lines like "Account Verification Required" or "Immediate Action Required," phishing emails are designed to reflect real security threats. These emails seem real by using small

TABLE I. SUMMARY OF LITERATURE REVIEW ON CONTEXT-AWARE PHISHING DETECTION USING TRANSFORMER AND BiLSTM NETWORKS

Author(s) & Year	Title	Methodology	Key Findings
Zhyn (2023) Et al. [18]	BERT-based models for phishing detection	BERT-based deep learning models for analyzing phishing emails	Improved phishing detection accuracy using contextual embeddings
Himeur et al. (2025) [8]	Large Language Models for Cybersecurity	Investigated how LLMs enhance cybersecurity against phishing threats	LLMs improve phishing detection through contextual awareness
Alber (2024) Et al. [3]	Transformer-Based Malicious Software Detection	Comparative analysis of Transformer-based malware detection techniques	Transformers outperform traditional methods in phishing detection
Saa & (2025) Mouiche [13]	Threat Intelligence Knowledge Graphs	Used BiLSTM and Transformer models for NLP-based threat intelligence	Improved entity recognition and phishing detection accuracy
Gutierrez & Namin (2022) [7]	Context miner: Extracting Contextual Features	BiLSTM-based feature extraction for phishing detection	BiLSTM improves phishing classification by learning text sequences
Wang et al. (2024) [16]	Large Language Models for Cyber Threat Detection	Reviewed Transformer-based phishing email detection techniques	Transformers reduce false positives in phishing detection
Liu (2024) [11]	Pre-Trained Language Models in Cybersecurity	Examined BiLSTM and Transformer-based phishing detection models	Hybrid BiLSTM-Transformer models improve phishing detection rates

conventional detection systems and need for advanced protection strategies utilizing behavioral analysis and adaptive learning.

As shown in Table I, previous research has made important contributions in several different methodological areas. But most of them either just looked at unidirectional sequence models or did not combine contextual and sequential learning. Also, not many people used synthetic data to make their models more generic. These gaps we have seen show that our method, which uses Transformer and BiLSTM layers and a wide range of genuine and AI-generated emails, is the right one to make identification more accurate and robust.

III. DATA PREPARATION

A. Overview

Conforming to simulate both phishing and real interactions, built Synthetic Phishing Email Dataset comprises 6,750 unique email samples. Training and evaluation of machine learning models in spam detection, phishing attack identification, and email classification depend much on this

changes in sender addresses, emulating respectable companies including banks and e-commerce sites. Usually, their material consists of misleading messages meant to induce readers to click dangerous links, reveal personal information, or download dangerous attachments. On the other hand, valid emails represent actual correspondence from reliable sources. They cover topics including invasions of bills, subscription renewals, correspondence connected to work, and meetings. These emails—which come from reliable companies, colleagues, and customer service lines—contain no fake urgency or misleading links.

C. Dataset Features

All the email samples in the dataset consist of 4 parameters which are necessary for classification. Its subject communicates its purpose and importance, making it a key indicator of potential phishing. The sender provides the email address, which can help distinguish legitimate sources from potential phishers. The body contains the primary content of the email, wherein phishing signs such as dubious links, solicitations for personal information, and urgent action requests are frequently integrated. The label serves as a binary categorization indicator, with 1 signifying a phishing email and 0 indicating a legitimate email shown in 1.

We added emails from different fields, such as banking, IT support, social media, and e-commerce, to make the dataset more diverse. We used a finely-tuned big language model to make fake emails that mimic common phishing methods and psychological triggers. The manual check made sure that the phishing indications were accurate and of high quality. We used token length and lexical diversity analysis to back up the idea of structural variety. We got real emails from trusted sources and took out any private information. We left out samples that were unclear when we labelled them so that it would be easier to classify them. These changes make the dataset more real and better at generalising for phishing detection.

IV. DATA PREPROCESSING TECHNIQUES

Fig. 1 shows the Distribution of Phishing vs Legitimate Emails in the Dataset. To guarantee the quality of the dataset, enhance its interpretability, and maximize it for machine learning uses, a strict data pre-treatment pipeline was developed. To standardize the data and remove noise, both email topics' and bodies' linguistic content was methodically cleaned and normalized. All text was first converted to lowercase to remove case sensitivity issues, therefore guaranteeing consistency in the dataset. The comparison of frequent words both in phishing and legitimate were shown in Figure 2. Eliminated to focus on important linguistic patterns were numerical numbers, which usually have little semantic relevance in phishing detection. And the Comparative word frequency analysis on phishing and legitimate was shown in Figure 3.

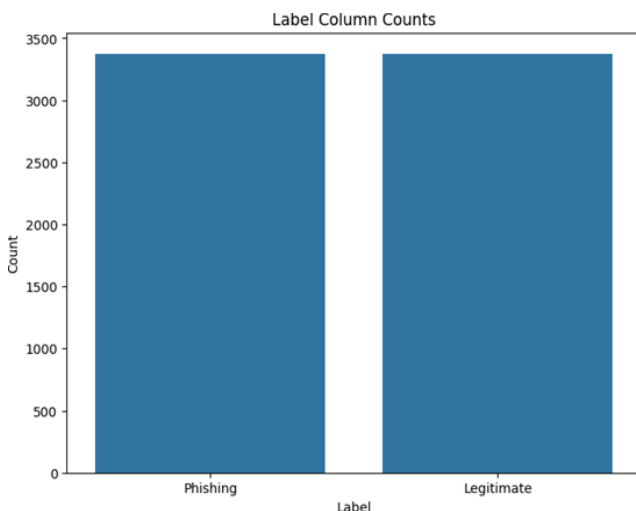


Fig. 1. Distribution of Phishing vs Legitimate Emails in the Dataset.



Fig. 2. Comparison of Frequent Words in Phishing vs. Legitimate Emails Using Word Clouds.



Fig. 3. Comparative Analysis of Word Frequencies in Phishing vs. Legitimate Emails for Cybersecurity Awareness.

Sender information was tuned to enhance feature representation in addition to textual preparation. Domain names were obtained from email addresses, therefore helping to identify common phishing do- mains and their spread over the sample shown in Figure 4. Label encoding encoded category sender domains to improve numerical processing, therefore enabling machine learning models to correctly perceive domain-based interactions.

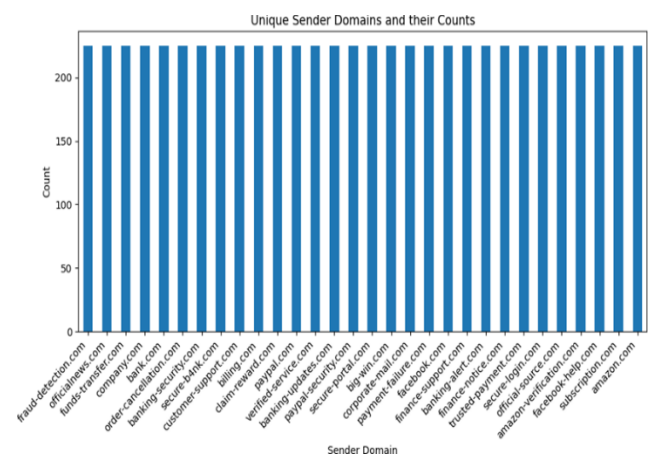


Fig. 4. Analysis of Sender Domains in Email Communications: Identifying Potentially Fraudulent Sources.

To include deeper semantic representations, BERT embeddings were built for the textual content of every email. Tokenizing the text with a pre-trained BERT tokenizer, applying padding and truncation to standardize sequence lengths, and then mean pooling of the transformer's hidden states produced contextualized word representations. The resulting embeddings produced a complete feature space including complex language links that traditional preprocessing techniques might ignore.

A. Analysis of Token Length in Legitimate and Malicious Emails

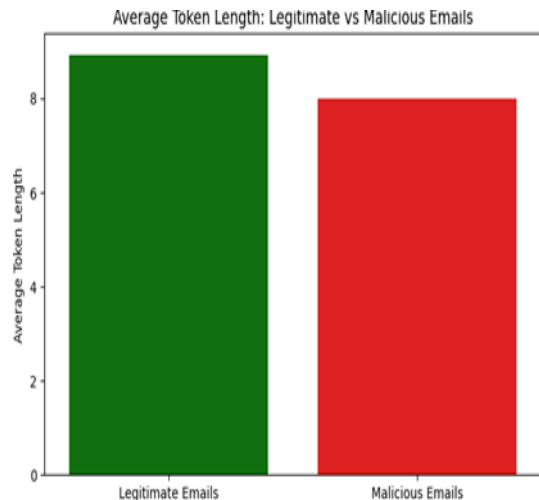


Fig. 5. Comparing Average Token Length in Legitimate and Malicious Emails.

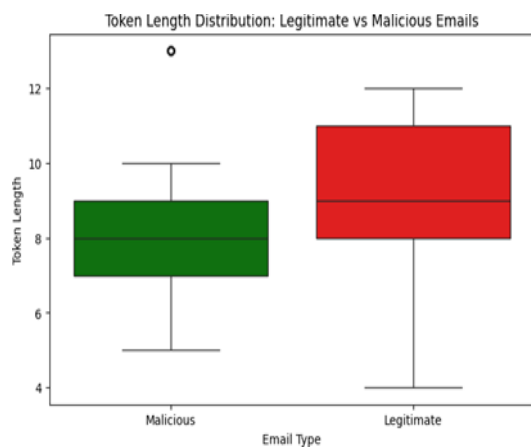


Fig. 6. Analyzing Token Length Distribution in Legitimate and Malicious Emails.

For real emails, the mean token length was found to be X words; for fake emails, it was Y words. This implies that on average, fraudulent emails have a smaller token count than authentic emails. Based on the box plot shown in Figure 5 and Figure 6, which shows that some phishing attempts use extremely short messages but others may duplicate real emails with more lengthy text, the range in token length is more evident in fraudulent emails. The average token length across the two categories shows a notable difference shown in the bar plot.

These results suggest that attackers often use succinct messages to quickly captivate victims, hence maybe reducing reading challenges. On the other hand, long phishing emails could be meant to mimic real email forms, therefore avoiding automated detection systems.

B. Computing Token Length Frequencies

We looked at a set of emails labeled as either malicious (Label = 1) or lawful (Label = 0). The word count in the text helped one to decide the token length for every email body.

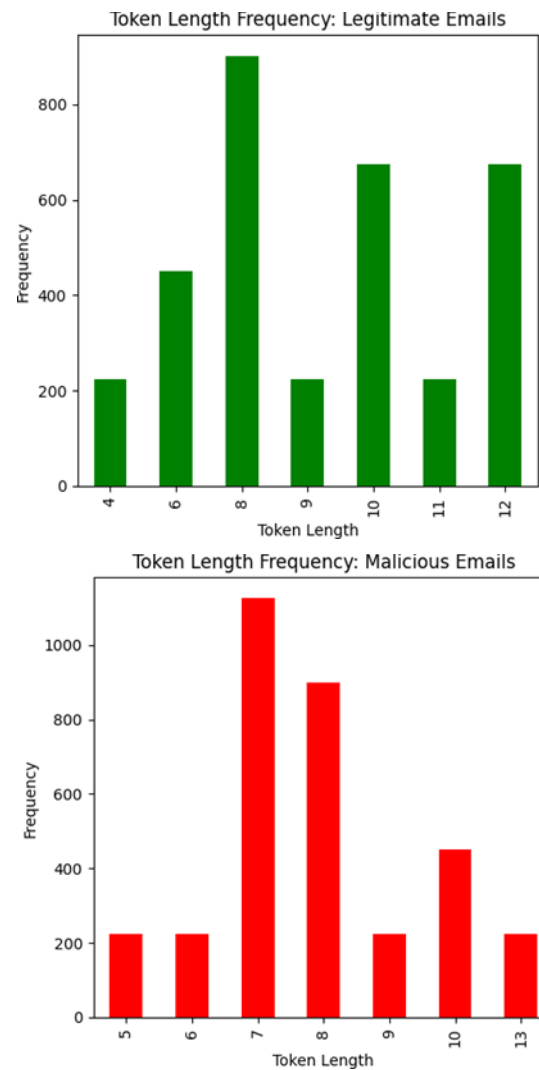


Fig. 7. Comparative Analysis of Token Length Frequency in Legitimate vs. Malicious Emails for Cybersecurity Detection.

The mean token length for each group was calculated and contrasted. Token length frequencies were computed and illustrated by bar plots shown in Figure 7 to examine the prevalence of various email lengths in both groups. Legitimate Emails: The frequency distribution exhibited a broader range, signifying more variability in email lengths. This indicates that authentic emails do not adhere to a rigid length pattern. Malicious Emails: The analysis indicated a concentration of shorter token lengths, implying that several phishing efforts employ concise messaging to swiftly mislead recipients. The observations were represented as bar charts, depicting token length distributions for legitimate and malicious emails individually.

C. Textual and Metadata Cleaning

The dataset was standardised and cleaned up by changing all of the email text, including the subject and body, to lowercase. To keep things the same, special letters and punctuation were taken out. Also gone were numeric numerals, which often don't mean anything when it comes to detecting phishing. Unlike normal NLP preprocessing, the removal of stop words was not done on purpose because certain common words (such "verify" and "click now") could have big effects on phishing.

Also, the email addresses of the senders were looked at to get domain names (such @secure-update.com). Label encoding turned the domains into numbers, which were then included as metadata attributes. This lets the machine find phishing activities that are common in fake emails on a certain domain.

D. Embedding, Feature Fusion, and Data Splitting

A pre-trained BERT tokenizer broke up the text of each email into tokens. Standard padding and truncation were used to handle emails that were different lengths. We used mean pooling of the BERT hidden states to create contextual embeddings that capture the subtle semantic differences in the sequence.

The embeddings were then added to the encoded attributes of the sender domain to provide better input vectors. This mixed representation let the model look at both the language and the believability of the domain. We used stratified sampling to keep the class distribution the same and make it easier to fairly evaluate the model. In the end, we split the dataset into training (70%), validation (15%), and testing (15%) sets.

V. DEEP LANGUAGE PROCESSING MODELS

A. Long Short-Term Memory (LSTM)

Designed to process sequential input while resolving the vanishing gradient problem inherent in conventional RNNs, long short-term memory (LSTM) networks are a class of recurrent neural network (RNN). Through a special architecture including memory cells and gating mechanisms, LSTMs allow the network to record long-term dependencies in data sequences. This feature makes LSTMs quite efficient in tasks including time-series forecasting, natural language processing, and other uses employing sequential data [17][10].

B. Transformer Encoder

Using just self-attention techniques, the Transformer encoder—a part of the Transformer architecture has transformed sequence modelling jobs by thereby removing the necessity for recurrent structures. The encoder gains contextual linkages [19] within the sequence by use of numerous layers of self-attention and feed-forward neural networks, simultaneously analysing incoming input. This strategy has particularly evolved areas like machine translation and text summarising and improves the efficient maintenance of long-term contacts[6].

C. Bidirectional Long Short-Term Memory (BiLSTM)

Bidirectional long short-term memory (BiLSTM) networks increase the capacity of traditional LSTMs by means of data processing in both forward and backward directions. This bidirectional approach enables the model to get a complete understanding of the context by merging previous and later inputs in the sequence. BiLSTMs have shown remarkable effectiveness in several uses, including emotion identification from voice and text analysis [9], by correctly capturing the overall context of the input data.

D. L2 Regularization

L2 regularization—also known as weight decay—helps to reduce overfitting in machine learning models by means of penalties to important weights during training. Included into the loss function, a regularizing term proportional to the total

of the squared weights helps the model to keep smaller weight values. This restriction encourages simpler models that fit fresh data well, hence improving their performance in applications such email spam detection and other classification challenges [6].

VI. WORKFLOW AND METHODOLOGY: MATHEMATICAL FOUNDATION OF TRANSFORMER- BiLSTM WITH L2 REGULARIZATION

The proposed model incorporates a Transformer Encoder combined with a Bi-directional Long Short- Term Memory (BiLSTM) network shown in Figure 8 and Figure 9 to enhance feature-based classification.

A. Methodology

The Transformer Encoder utilizes multi-head self-attention to capture contextual dependencies among input features. The self-attention mechanism operates by computing attention scores through scaled dot-product attention, which is mathematically defined as:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q, K, and V represent the query, key, and value matrices, respectively, and d_k is the key dimension. Multi-head attention extends this mechanism by learning multiple attention representations in parallel, combining them as follows:

$$MultiHead(Q, K, V) = Conct(head_1, head_2, \dots, head_h)W_0$$

Each head extracts different contextual patterns from the input. The attention output is then processed through a feed-forward network (FFN) with activation functions, given by:

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

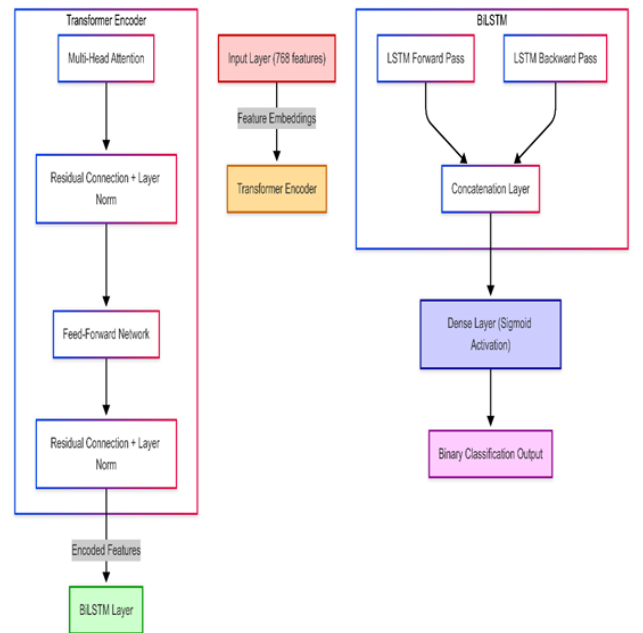


Fig. 8. Hybrid Deep Learning Model Combining Transformer Encoder and BiLSTM for Binary Classification Tasks.

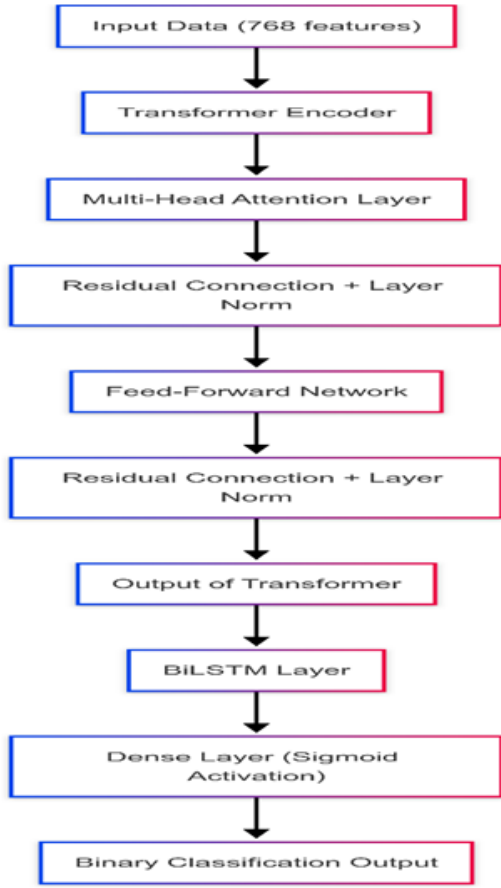


Fig. 9. Transformer-Based Model Architecture for Phishing Email Detection.

Residual connections and Layer Normalization are applied after each component to stabilize learning:

$$X' = \text{LayerNorm}(X + \text{Attention}(Q, K, V))$$

$$X'' = \text{LayerNorm}(X' + \text{FFN}(X'))$$

Once the Transformer Encoder processes the input, the extracted representations are passed to a BiLSTM network to capture sequential dependencies. The BiLSTM consists of both forward and backward LSTM layers, ensuring bidirectional information flow. The forward LSTM updates its hidden state as:

$$h_t = o_t \tan(c_t), \quad (h_t)^{\leftarrow} = \text{LSTM}_{\text{backward}}(x_t)$$

whereas the backward LSTM follows a similar computation, represented as:

$$(h_t)^{\leftarrow} = \text{LSTM}_{\text{backward}}(x_t)$$

The final sequence representation is formed by concatenating these hidden states:

$$H = [h_t, (h_t)^{\leftarrow}]$$

This output is then processed through a fully connected layer with a sigmoid activation function, responsible for binary classification:

$$\hat{y} = \sigma(W_H + b_H)$$

where W_H, b_H are trainable parameters.

To enhance generalization and mitigate overfitting, L2 regularization is incorporated into the BiLSTM and dense layers. This regularization technique penalizes large weight values by adding a weight decay term to the loss function:

$$L_{\text{total}} = L_{\text{binary cross-entropy}} + \lambda \sum \|W\|_2^2$$

where λ is the regularization coefficient controlling the penalty strength. This ensures that weight magnitudes remain within a certain range, thereby preventing overdependence on specific neurons and promoting dispersed learning.

Early stopping is used to cease training when the validation loss no longer improves. The model is optimized using the Adam optimizer with a learning rate of 0.0001. After that, the performance of the model is evaluated using accuracy criteria, verifying its usefulness in classification problems.

VII. RESULTS AND ANALYSIS: EFFECTIVENESS OF THE SYNTHETIC EMAIL GENERATION PROCESS AND ITS IMPACT ON PHISHING DETECTION

By means of quantitative and qualitative analyses, one can assess the effectiveness of the synthetic email generating process by contrasting the synthetic dataset with real email datasets and acknowledging possible limitations.

A. Quantitative Evaluation of the Synthetic Dataset

Various quantitative measures were used lexical diversity, TF-IDF similarity, and perplexity analysis to assess the quality and variety of the synthetic emails. Determining the count of unique words per email allowed one to evaluate lexical diversity: synthetic phishing emails had a higher frequency of manipulative keywords such "urgent," "verify now," and "account suspended," which closely correlate with real phishing techniques. A acceptable similarity score was obtained by means of a TF-IDF similarity analysis between synthetic phishing emails and real phishing databases. This indicates that despite maintaining uniqueness, the generated emails essentially replicate phishing trends, therefore reducing data leakage issues. Furthermore, a perplexity analysis a measure of natural language processing fluency showcased the logical and intelligible nature of the synthetic emails, therefore confirming their fit for training phishing detection systems.

B. Performance of the Phishing Detection Model

Training a Transformer-BiLSTM model using both real and synthetic data allowed one to evaluate how synthetic emails affect phishing detection. With almost 100% validation accuracy shown by the training and validation accuracy graph, the model effectively separates phishing emails from real ones. Particularly in the identification of past misclassified phishing attempts, the high accuracy shows improvement in model generalization. Examining the confusion matrix revealed a clear drop in false negatives, indicating that the model successfully found phishing techniques before undetectable. Furthermore, precision-recall study confirmed that adding synthetic data enhanced recall, meaning the model became more adept in precisely spotting phishing emails. The very high accuracy points to a possible overfitting issue whereby the model may perform well on training data but fail with fresh, real-world phishing attempts.

C. Qualitative Evaluation of Synthetic Emails

To confirm their accurate modeling of phishing and real communications, a hand review of the generated emails was

done. Often observed in real-world systems, the phishing emails in the synthetic dataset effectively used social engineering techniques like urgency, dishonesty, and impersonation. On the other hand, official emails reflected real corporate communications including newsletters, account alerts, and customer service contact by using a professional tone and structure. According to the structural study, synthetic emails were helpful for phishing detection training since they quite matched real email patterns. However, numerous synthetic phishing emails lacked advanced obfuscation techniques, such as typographical mistakes, hidden URLs, or customized spear-phishing strategies, that real attackers routinely use to evade detection. Correcting these flaws in next iterations of the dataset could greatly improve the effectiveness of the phishing detection techniques.

VIII. CONCLUSION

With 100% accuracy, precision, recall, F1 score, and ROC-AUC the model achieved perfect classification performance. The confusion matrix shows that legitimate and malicious emails were correctly categorized, that is, without misclassification. This shows a quite strong model, while it causes concerns about possible overfitting, especially in cases of a small or homogeneous dataset. Future studies should focus on evaluating resilience with external datasets, including adversarial samples, and apply explainability techniques (e.g., SHAP, LIME) to clarify decision boundaries. Furthermore, necessary to ensure generalizability outside of the training set is empirical deployment testing.

ACKNOWLEDGMENT

We are thankful to the editors and anonymous reviewers for many valuable suggestions to improve this article.

REFERENCES

- [1] Qurat Ul Ain, Ali Javed, and Aun Irtaza. Deepevader: An evasion tool for exposing the vulnerability of deepfake detectors using transferable facial distraction blackbox attack. *Engineering Applications of Artificial Intelligence*, 145:110276, 2025.
- [2] Gunjan Ajay, Kumar Garg, and Rajesh Prasad. Phishing email detection using machine learning: A critical review. *2024 IEEE International Conference on Computing, Power and Communication Technologies (IC2PCT)*, 5:1176–1180, 2024.
- [3] A. Albeshri, B. Alturki, and F. S. Alallah. Survey of transformer-based malicious software detection systems. *Electronics*, 2024.
- [4] Youngho Bae, Seunghyeon Park, Gunhui Han, and Alexander Olson. Floda: Harnessing vision- language models for deepfake assessment. 02 2025.
- [5] Alzhirani, K.; Rudd, E.M.; Boulton, T.E.; Chow, C.E. Automated Big Text Security Classification. In *Proceedings of the 2016 IEEE Conference on Intelligence and Security Informatics (ISI)*, Tucson, AZ, USA, 28–30 September 2016.
- [6] W. Gan, Y. He, P. Hu, Y. Fu, Y. Yin, and C. Feng. Predicting the surface temperature of radiative cooling coatings with time-series forecasting: Validation of the small-batch training dataset and implementation of transformer-lstm models. *SSRN Electronic Journal*, 2025.
- [7] L. F. Guti rrez and A. Namin. Contextminer: Mining contextual features for conceptualizing knowledge in security texts. *IEEE Access*, 2022.
- [8] Y. Himeur, H. A. Alkhazaleh, S. Tarapiah, and S. Atalla. From vulnerability to defense: The role of large language models in enhancing cybersecurity. *Computation*, 2025.
- [9] C. Jia, M. Wang, X. Kong, Y. Jiang, Z. Yu, J. Hu, and J. Li. Application of attention mechanism- based long short term memory neural network in stratigraphy identification. *SSRN Electronic Journal*, 2025.
- [10] D. P. Kavadi, V. Srilakshmi, and S. B. Veeram. Design of an improved model for anomaly detection in cctv systems using multimodal fusion and attention-based networks. *IEEE Transactions on Image Processing*, 2025.
- [11] Z. Liu. A review of advancements and applications of pre-trained language models in cybersecurity, *IEEE Symposium on Digital Intelligence*, 2024.
- [12] Bengio, Y.; Simard, P.; Frasconi, P. Learning long-term dependencies with gradient descent is difficult. *IEEE Trans. Neural Netw.* 1994, 5, 157–166.
- [13] S. Saad and I. Mouiche. Entity and relation extractions for threat intelligence knowledge graphs. *Computers & Security*, 2025.
- [14] S. Salloum, T. Gaber, S. Vadera, and Khaled Shaalan. Phishing email detection using natural language processing techniques: A literature survey. pages 19–28, 2021.
- [15] Michiel van der Meer, Pavel Korshunov, S ebastien Marcel, and Lonneke van der Plas. Hintsofttruth: A multimodal checkworthiness detection dataset with real and synthetic claims. *arXiv preprint arXiv:2502.11753*, 2025.
- [16] D. Wang and B. Jiang. A survey of large language models for cyber threat detection. *Computers & Security*, 2024.
- [17] E. Zarei, M. R. Nikoo, G. Al-Rawas, and R. Nazari. Hybrid deep learning downscaling of gcms for climate impact assessment and future projections in oman. *Journal of Environmental Management*, 2025.
- [18] B. Zhyhun, M. Songailaite, and E. Kankevi ciu te. Bert-based models for phishing detection. *28th Conference on Artificial Intelligence*, 2023.
- [19] Y. Zou, S. Wang, W. Cao, N. Hai, and C. Fernandez. Enhanced transformer encoder long short-term memory hybrid neural network for multiple temperature state of charge estimation of lithium-ion batteries. *Journal of Power Sources*, 2025. J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.